Dynamic Sensing and Correlation Loss Detector for Small Object Detection in Remote Sensing Images

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***Abstract*— Recently, significant object detection achievements have emerged for optical remote sensing images. However, the performance and efficiency of small object detection are still highly unsatisfactory because of the scale diversity between the objects; furthermore, small objects always have small amounts of effective information, which are difficult to locate. To address this problem, we propose a novel dynamic sensing and correlation loss detector (DCDet) for performing object detection in remote sensing images. The detector consists of two modules: a small object dynamic sensing (SODS) module and a simple but effective correlation loss (CrLoss) function. SODS is utilized to capture the information of small objects in a scale sequence. We consider the feature pyramid as a set of video frames when the camera is zoomed in on the image and use the object focusing module in dynamic sensing to always focus on the small objects in each video frame. The detection performance achieved for small objects is improved by shifting the detector's attention from the entire image to small objects within the frame to provide a multiscale feature representation of the small objects and their contextual information. The CrLoss is a special CrLoss for remote sensing image object detection tasks and directly optimizes the correlation coefficient to improve the performance of a detector. Extensive experiments conducted on the publicly available DOTA, DIOR-R, and HRSC2016 datasets show that our DCDet outperforms the existing state-of-the-art remote sensing object detection methods in terms of many evaluation metrics.**

***Index Terms*— Dynamic sensing, remote sensing images, small object detection.**

NOMENCLATURE

$\boldsymbol{P}_l$	Scale feature at level l.
$\widetilde{\boldsymbol{P}}_l$	After strengthening the scale characteristics of l level.
$\mathbb{R}$	Set of real numbers.
N	Number of classes for objects.
$\boldsymbol{s}_{\text{pre}}$	Confidence score of ith object proposal r.
$\boldsymbol{\alpha}$	Horizontal box for a 4-D vector.
$[x, y, w, h, \theta]$	Center point coordinate, width, high, and angle of ith object proposal r.
$[H, W, C]$	Width, high, and channel of features.
$\boldsymbol{g}_i^*$	True label of the ith anchor.
$\boldsymbol{s}_i^p$	Offset of the predicted bounding box with respect to the ith anchor.
$\boldsymbol{s}_i^g$	Offset of the ground truth with respect to the ith anchor.
$W_{\text{cls}}, w_{\text{reg}}, w_{\text{corr}}$	Contribution weight of classification, regression, and correlation loss.

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Data is available on-line at https://github.com/CHaunceyshen/DCDet.

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I. INTRODUCTION

WITH the continuous development of remote sensing technology and improvements achieved by satellite systems, high-resolution remote sensing images can be easily acquired. However, remote sensing images contain many small objects, such as vehicles, buildings, and airplanes; these objects can be occluded within complex backgrounds and are difficult to visually recognize, especially in large-scale remote sensing images. Deep learning-based object detection algorithms have been widely used and continuously studied [15], [17], [18], [19], [20], [21], [22], and some progress has been made in terms of their performance. However, small object detection is still a challenging problem, especially in large-scale remote sensing images.

Recently, several methods have demonstrated the importance of leveraging rich contextual information for detecting visually challenging small objects. Liu et al. [15] designed a feature pyramid network (FPN) based on a two-stage approach that combines high-resolution feature maps with low-resolution feature maps and assigns them to independent pyramid levels based on object sizes. To improve the performance of the FPN, various methods that fuse pyramid features have been proposed [15], [46], [47], [48], which utilize models better suited for small object detection. However, when FPNs or other detection models address multiscale features, they simply use the fusion methods of upsampling and downsampling and then summation or concatenation. They spread the attention of the utilized model to all the objects in the whole image without focusing on the different information exhibited by small objects at different scales, especially the same object across multiple scales. Thus, small objects have serious interferences from the information of large objects

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Fig. 1. Illustration of the difficulty of the small object detection task. The left image shows a large remote sensing image, and the red window shows the size of the cropped image produced during training; here, it can be observed that detecting densely distributed small objects is very difficult. The right image shows the result of an image being detected after implementing a simple multiscale feature fusion operation, and the small car next to the harbor is ignored by the detector.

when processing a feature map. As shown in Fig. 1, small objects located next to large objects are difficult to detect.

Most object detectors optimize the weighted sum of the classification and regression losses during training. Recent findings [53], [54], [54] have shown that the performance can be improved when these two loss functions are forced to interact. However, these methods require the addition of extra auxiliary heads, which burdens the constructed model and is clearly unsuitable for addressing high-resolution object detectors for remotely sensed images. Because the burden imposed on an object detector by remote sensing images is already high, adding another auxiliary head greatly increases the training and inference times of the model.

To address the aforementioned issues, a dynamic sensing and correlation loss detector (DCDet) is proposed. The network consists of two key modules: a novel SODS module and a new plug-and-play classification-regression correlation loss (CrLoss) function. We regard the feature maps of an FPN as a sequence of temporal video frames and use the object focusing module in SODS to focus attention on the small objects in the video frames for providing multiscale feature representations of the small objects and contextual information. Furthermore, we design a new CrLoss function to strengthen the correlation between the classification and regression tasks. We can improve the detection performance of the model by directly optimizing the correlation coefficient. This solution does not require additional auxiliary heads and does not affect the size, inference time, or training time of the model. In addition, CrLoss is applicable to all remote sensing image object detectors and is a plug-and-play CrLoss function.

The main contributions of this article are given as follows.

1) We propose a new SODS module for small object detection. We consider a pyramid feature map as a set of time-series video frames and utilize the object focusing module in dynamic sensing to focus the model's attention on small objects, providing richer multiscale feature representations for small object detection.
2) We design a new plug-and-play CrLoss function between the classification and regression tasks; CrLoss can effectively improve the performance of the detector by optimizing the correlation coefficient.
3) We compare our method with other existing object detectors for remote sensing images. Our DCDet is thoroughly validated on three public datasets DOTA, DIOR-R, and HRSC2016. In addition, we evaluate our CrLoss on nine current popular detectors.

The rest of the article is organized as follows. In Section II, we introduce the related work. Section III presents a detailed description of our method. Section IV introduces the loss function used in our method. Section V introduces the details of our experiments. Finally, in Section VI, we conclude our findings.

II. Related Works

Here, we discuss two main types of approaches: region-based (two-stage) methods and regression-based (one-stage) methods. Furthermore, we introduce representative work on small object detection and research advances on the CrLoss between classification and regression tasks.

A. One-Stage Object Detection

One-stage detection methods have gained significant attention due to their superior real-time performance. For example, Guo et al. [7] proposed a one-stage network with a correlated feature alignment module (CFAM) to address the problem of rotation object detection. Qian et al. [8] adjusted the rotation loss to eliminate the issue of loss discontinuity, resolving the unstable training and performance degradation challenges faced by previous methods. Xu et al. [9] addressed the spatial and scale mismatches between the predefined anchors and the features extracted by convolution in one-stage detectors by proposing an efficient feature alignment method. Li et al. [10] introduced an instance-aware distillation method to strengthen the relationships between the different instances. To reduce the influences of noise and blur, Yi et al. [12] proposed an object saliency enhancement strategy that separates the foreground and background. Yang et al. [13] achieved rapid and accurate object detection by using a progressive regression approach from coarse granularity to fine granularity.

B. Two-Stage Object Detection

In contrast to the former approaches, two-stage detection methods have been widely proposed and developed due to their high precision. For instance, considering the unique spatial distributions of objects in remote sensing images, Yu and Ji [2] proposed a two-stage remote sensing image object detector that used deformable convolutions to adapt to the geometric variations exhibited by different object instances. Wang et al. [3] introduced an end-to-end feature feedback pyramid network to address object scale and category differences. Li et al. [4] presented a novel feature attention-based object detection framework that focused on learning intrinsic representations of objects in an end-to-end framework. Yang et al. [5] innovatively incorporated denoising concepts into object detection to enhance the detection effect achieved

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for small and cluttered objects by performing instance-level denoising on feature maps. Tian et al. [6] proposed a long short-term memory- and relation-enhanced embedded graph attention network to address the problem caused by neglecting the relationships between objects.

C. Small Object Detection

Small object detection is a challenging task in computer vision. To improve the effectiveness of small object detection, Lin et al. [14] proposed the FPN structure, which combines high-resolution feature maps with low-resolution feature maps. For example, large objects are detected in a low-resolution pyramid feature map, while small objects are detected in a high-resolution pyramid feature map. Various FPN-based models have been proposed to alleviate the semantic gap between the different layers of pyramid feature maps [4], [15], [45], [46], [47].

For instance, Zhang et al. [49] designed a network that included a deconvolution layer after the last convolutional layer of the base detector for small object detection. Yuan and Zhang [50] introduced a low-coupling robust regression module to improve the precision of object localization and prevent the loss of small objects. Courtrai et al. [51] considered the number of network layers and utilized deep residual blocks integrated into Wasserstein generative adversarial networks (GANs) to learn a superresolution network. Ahmad et al. [52] enhanced the small-scale features of an FPN by combining a monocular detector with an image pyramid network to extract semantically strong features and generate candidate regions. Li et al. [4] proposed a novel object detector based on GANs to simultaneously detect multiple classes of small objects.

D. Correlation Between Classification and Localization

Recent object detectors have attempted to predict consistent outputs for two independent tasks by focusing on the centers of objects [19], [53], [54]. The anchor points of object centers are presumed to provide more accurate predictions for classification and localization purposes. For example, Tian et al. [19] used additional auxiliary heads during training to supervise the intersection over union (IoU), mask IoU, or uncertainty metrics based on the localization quality of the detection result, such as the center point. Feng et al. [54] achieved a better balance between task interactions and task-specific features through a task alignment head (T-Head).

The existence of a large number of small objects and diverse and complex scenes is a great challenge encountered when performing object detection in high-resolution remote sensing images. The use of simple up-and-down sampling and fusion for the multiscale features to achieve enhanced small object detection does not produce satisfactory results. Furthermore, due to the high resolution of remote sensing images, the existing correlation guidance that requires the addition of additional auxiliary heads is not applicable to object detection in remote sensing images. To address the above issues, we propose a dynamic perception module that specializes in small object detection and design a new plug-and-play correlation solution. This solution is applicable to all object detectors, both NMS-based and NMS-free detectors.

III. Methodology

Let $\boldsymbol{F} = \{\boldsymbol{P}_l \in \mathbb{R}^{|(H/2_l)|\times|(W/2_l)|\times C}\}$ with $\boldsymbol{l} \in \{2, 3, 4, 5, 6\}$ denoting the multiscale feature maps obtained by an FPN, and let $\boldsymbol{P}_l$ denote the feature at a single scale. The multiscale features enhanced by SODS are denoted by $\boldsymbol{F}'$. The class confidence scores predicted by the proposed approach are labeled $S_{\text{pre}} = \{s_1, \ldots, s_N\} \in \{0, 1\}^N$, where N denotes the number of classes. $[x, y, w, h, \theta]$ denotes the five parameters used for box detection. In nomenclature, we present a summary of the notations used by our approach.

A. Framework

In this section, we introduce a novel remote sensing image detector: a DCDet. It consists of five parts: a backbone network with an FPN for generating multiscale feature maps, a dynamic sensing module for enhancing the feature information of small objects, two detection heads for classification and regression tasks, and a CrLoss for supervising the classification and regression tasks, as shown in Fig. 2. We adopt ResNet as the backbone architecture. We remove the fifth layer of ResNet, as excessively deep networks hinder the extraction of small objects.

If the input image size is $\boldsymbol{H} \times \boldsymbol{W}$, then the size of the feature map obtained by the FPN is $\boldsymbol{F}$, which denotes the level of the pyramid. The features $\boldsymbol{F}$ are input into the SODS module to obtain the enhanced multiscale features, which are denoted as $\boldsymbol{F}'$. For each spatial position in the feature maps of all pyramid levels, we assign three horizontal boxes with different aspect ratios of {1:2, 1:1, 2:1}. Each horizontal box $\boldsymbol{\alpha}$ is represented by a 4-D vector $\boldsymbol{\alpha} = (\boldsymbol{\alpha}_x, \boldsymbol{\alpha}_y, \boldsymbol{\alpha}_w, \boldsymbol{\alpha}_h)$, where $(\boldsymbol{\alpha}_x, \boldsymbol{\alpha}_y)$ denotes the center coordinates of the box and $\boldsymbol{\alpha}_w$ and $\boldsymbol{\alpha}_h$ denote the width and height of the box, respectively. $\boldsymbol{\alpha}$ and $\boldsymbol{F}'$ are input into the detection head to obtain classification and regression results. Notably, since our remote sensing image object detection process is directional, the parameters of the regression box are $\text{Box} = [x, y, w, h, \theta]$, where θ denotes the angle at which the regression frame is offset, as shown in Fig. 3.

B. SODS Module

The widely adopted summation and joining methods for integrating multiscale features do not enhance the ability of a model to effectively detect small objects in remote sensing images. These methods spread their attention over the entire input image, resulting in the interference of the small object information with the large-object information and the erosion of the small object information by the complex background information. To fully extract the detailed features and detailed location information of small objects from the multiscale features of remote sensing images, we propose an SODS module, as shown in Fig. 2.

In the feature maps of the feature pyramid, the generated feature maps exhibit a zooming-in effect similar to that of a camera in a video; this provides progressively detailed representations of the objects. During the process of advancing the feature map from a low resolution to a high resolution, the semantic information of the small objects gradually comes

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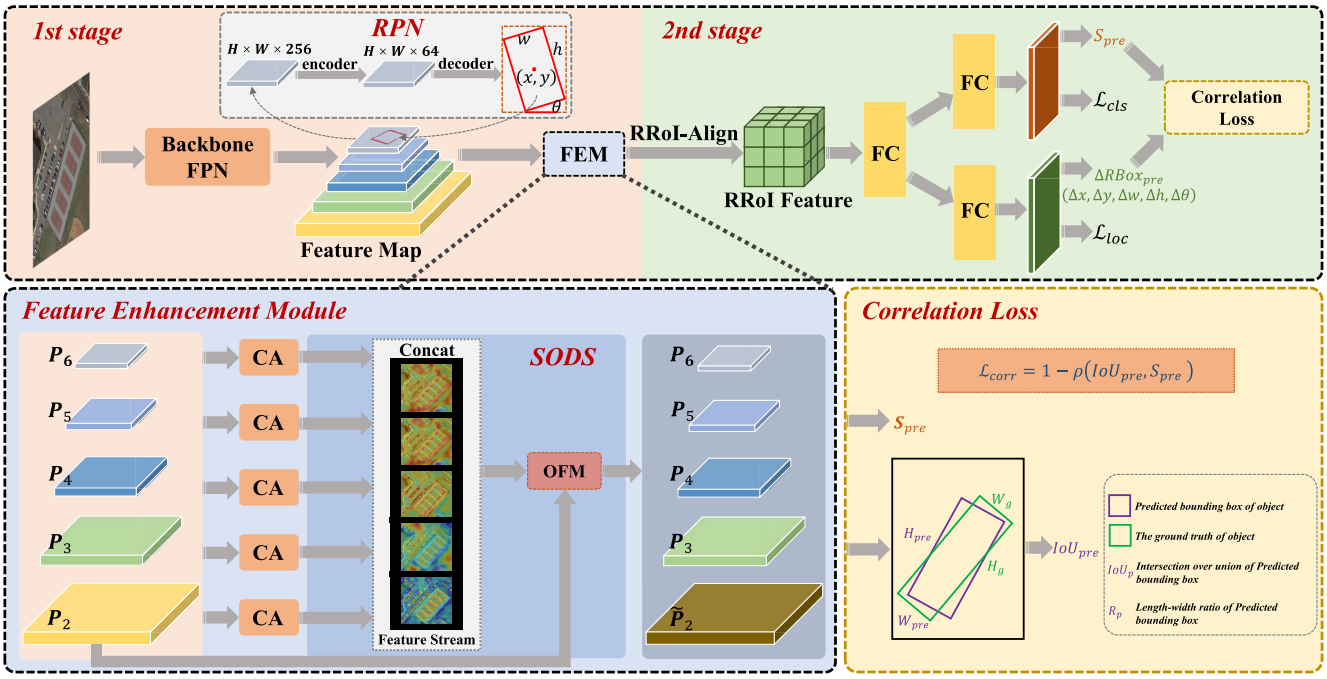


Fig. 2. DCDet network framework. The backbone and FPN are utilized for feature extraction in the remote sensing image analysis tasks. The SODS module extends the multiscale features of the 2-D images into 3-D space to form a new scale sequence. We focus on the changing features of the object in the scale sequence via 3-D convolution to better recognize and localize small objects. The CrLoss module is a simple and effective loss function designed specifically for remote sensing image object detection. This loss module provides improved model performance by associating the IoU and confidence scores to supervise the classification and regression tasks.

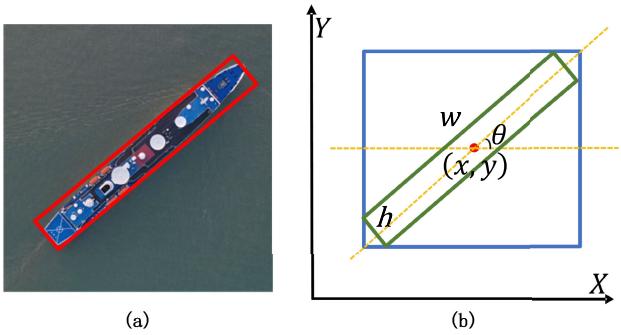


Fig. 3. Directional regression. (a) Example of a directed regression representation. (b) Schematic of the directed regression representation.

to the fore. This occurs because a higher spatial resolution can provide clearer and more explicit spatial information, and fine-grained texture and edge information are gradually enriched. Thus, the detailed features of the small objects can be better captured and represented. Therefore, by considering the pyramid feature map as a set of time-series video frames, we use the object focusing module to focus the attention of the model on the small objects in the video frames to better recognize and localize these objects. Due to this processing strategy, the model focuses on small objects all of the time, and the interference caused by large objects and complex background information is reduced to allow the semantic and spatial information of small objects to be gradually revealed in multiple feature map layers, thus enabling the capture of the change information exhibited by small objects.

We first pass all pyramid feature maps through a coordinate attention (CA) [22] module to extract the positional information of the objects. The CA module decomposes channel attention into two parallel 1-D feature encodings, effectively embedding spatial coordinate information into the generated feature maps, as shown in Fig. 4. Specifically, two 1-D global pooling operations are utilized to aggregate the input features into two separate feature maps along the vertical and horizontal directions. These two feature maps with directional information are then encoded into two attention maps; each map captures the long-range correlation in one spatial direction of the input feature map. Consequently, the positional information can be preserved in the generated attention maps. Finally, the two attention maps are multiplied and weighted onto the input feature map to emphasize the positional information of the objects. The final output $\boldsymbol{F}_c$ can be written as follows:

$$\boldsymbol{F}_c(i,j) = \boldsymbol{x}_c(i,j) \times \sigma\left(g_h\left(\delta\left(\boldsymbol{f}^h\right)\right)\right) \times \sigma\left(g_w\left(\delta\left(\boldsymbol{f}^w\right)\right)\right) \quad (1)$$

where c denotes the number of channels. δ is a nonlinear activation function. g_h and g_w are two 1×1 convolutional transformation functions. The features weighted by CA contain rich positional information on the objects and can assist subsequent regression tasks in better localizing the object. Moreover, this step also facilitates the rapid extraction of object information in the scale feature stream.

After enhancing the pyramid feature maps with the CA module, we resize them to the corresponding resolutions of $\boldsymbol{P}_2$ using nearest neighbor interpolation. Then, we use the unsqueeze function to add a temporal dimension to each feature level with dimensions of (time, w, h, c) and concatenate them along the new dimensions to form a new view, which

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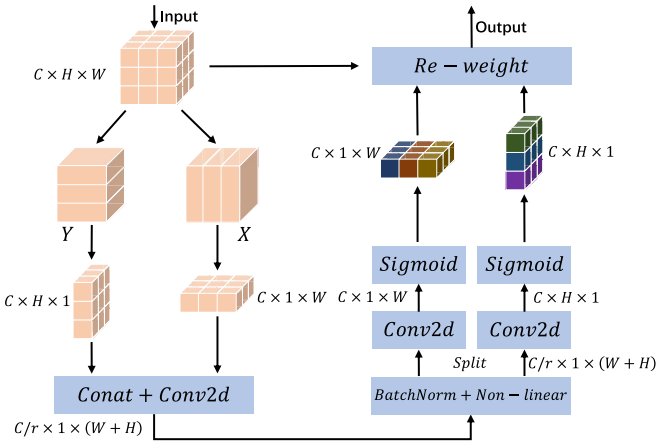


Fig. 4. CA module. By extracting attention maps embedded in the vertical and horizontal directions, we emphasize the positional information of objects at the individual scale.



Fig. 5. Object focusing module (OFM). The 3-D convolutions are utilized to extract the feature information of the small objects from the temporal video frames in the feature pyramid and obtain the output features $\widetilde{\boldsymbol{F}}_{\mathrm{DS}}$. They are then connected to $\boldsymbol{P}_2$, and a 1×1 convolution is applied to reduce the channel dimensionality.

is denoted as $\boldsymbol{F}_{\mathrm{DS}}$. This new view can be observed as a sequence of video frames, where each feature map represents a frame in the video stream. We then aggregate the small object information in the video frames using an object focusing module (OFM), which mainly consists of 3-D convolution, batch normalization, and ReLU activation functions. As shown in Fig. 5, the advantage of the utilized 3-D convolution over 2-D convolution is that the former can fully describe the relationships between the objects in 3-D space. The 3-D convolution can effectively utilize the scale information of the feature pyramid to gradually highlight the semantic and spatial information of small objects and enhance the representations of the texture and edge information. This can be expressed as follows:

$$\widetilde{\boldsymbol{F}}_{\mathrm{DS}} = g\big(\mathrm{ReLU}\big(\mathrm{Avg}\big(\boldsymbol{P}_i'\big)\big)\big) \tag{2}$$

where g is the 3-D convolution function. LeakyReLU is used as the ReLU function. $\boldsymbol{P}_i'$ is the feature scale obtained after resizing and expanding the dimensionality and can be expressed by the following formula:

$$\boldsymbol{P}_i' = \Theta(\Psi((\mathrm{CA}(\boldsymbol{P}_i)))) \tag{3}$$

where Θ is an extension of the unsqueeze function and Ψ denotes the nearest neighbor interpolation. We apply the obtained dynamic sensing features $\widetilde{\boldsymbol{F}}_{\mathrm{DS}}$ to the needed high-resolution feature maps since small objects are primarily detected on large-scale feature maps. However, the initial size of each dynamic sensing feature does not need to be in high resolution. The feature can be resized to different resolutions based on the size requirements of the object feature maps.

C. CrLoss Function

In object detection, we believe that two types of correlations are present between the classification task and the regression task: a semantic correlation and a spatial correlation. For the semantic correlation, the class information of an object provides important information regarding its attributes and shape and aids the regression task in accurately predicting the object's position and size. For spatial correlation, the object's position and bounding box information significantly impact the results of the classification task. Subsequently, the classification task's results can guide the regression task to better localize the object. Based on this, we propose a novel CrLoss function between the classification task and the regression task, and CrLoss includes consistency constraints between the classification and regression results to enhance their correlation. By introducing the CrLoss, the model can better coordinate the classification and regression tasks, thereby improving the overall performance of the object detection method.

Our CrLoss function supervises the classification task and the regression task while yielding improved model performance. The CrLoss is simply added using a weighting hyperparameter w_{corr}

$$\mathcal{L} = \mathcal{L}_{\mathrm{det}} + \omega_{\mathrm{corr}}\mathcal{L}_{\mathrm{corr}} \tag{4}$$

where $\mathcal{L}_{\mathrm{det}}$ is the basic loss function for object detection and consists of both a classification loss and a regression loss. The detailed formulation and explanation can be found in Section IV. $\mathcal{L}_{\mathrm{corr}}$ is the CrLoss function and is defined as follows:

$$\mathcal{L}_{\mathrm{corr}} = 1 - \rho\big(\mathrm{IoU}_{\mathrm{pre}}, \boldsymbol{S}_{\mathrm{pre}}\big) \tag{5}$$

where $\rho(\cdot)$ is a correlation coefficient. $\boldsymbol{S}_{\mathrm{pre}}$ and $\mathrm{IoU}_{\mathrm{pre}}$ are the confidence scores of the true and predicted categories and the IoU between the predicted bounding box and the labeled box, respectively. We optimized $\rho(\cdot)$ using Spearman's correlation coefficient; in addition, to solve the nondifferentiability problem encountered when the classified scores are sorted, we utilize the differentiable sorting operation of Torchsort. By converting the prediction results into ranks, the influences of outliers can be mitigated, and the correlation between the classification task and the regression task in the deep learning-based object detection paradigm can be more accurately measured.

This loss can simply be inserted into other detectors with classification and regression heads to improve the performance of the model. The resulting impacts on the size and inference speed of the model are negligible. On the other hand, the existing correlation methods usually focus on a single task, which not only introduces a large number of additional auxiliary computation heads but is also unsuitable for the task of object detection in remote sensing images.

IV. Overall Loss

To better train our model, we define an anchor as a positive sample if its IoU overlap with any ground-truth box is

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Algorithm 1 DCDet

Input: Ground-truth $\boldsymbol{G}$; Number of Labels $\boldsymbol{Y}$; Predefined anchor group $\boldsymbol{A}$; Reg, Cls and $\widetilde{\boldsymbol{F}}_{DS}$ are the regression results, classification results and feature in SODS; θ is the matching threshold.
Output: Multiclass object detection results.
for i in epochs **do**
  repeat
    Take a batch images M from trainval;
    Extract image feature by oriented R-CNN;
    Get FPN multi-scale features $\boldsymbol{P}_l$ by feature pyramid network;
    Use SODS Module obtain dynamic sensing feature $\widetilde{\boldsymbol{F}}_{DS}$;
    Generating new scale features $\widetilde{\boldsymbol{P}}_l$;
    for j in $\widetilde{\boldsymbol{P}}_l$ **do**
      Regressing the proposals on l yields the regression results;
      Classify the proposals of l and get the classification confidence scores;
      Calculate the IoU of the regression results with ground-truth;
      Backward propagate the loss and update the model parameters;
    End for
  Until all images have been fed into training models
End for

greater than 0.7 or if it has the highest IoU overlap with the ground-truth box and the IoU is above 0.3. Anchors with IoUs below 0.3 are considered negative samples. Anchors that do not satisfy both criteria are considered invalid samples and are disregarded during the training process. Importantly, the ground-truth box described above is the outer rectangle of the directional box enclosing the metrics.

Our loss function $\mathcal{L}$ is defined as follows:

$$\mathcal{L} = \frac{1}{N}\left(\sum_{i=1}^{N} w_{\text{cls}}\mathcal{L}_{\text{cls}}\left(\boldsymbol{p}_i, \boldsymbol{g}_i^*\right) + g_i^* \sum_{i=1}^{N} w_{\text{reg}}\mathcal{L}_{\text{reg}}\left(\boldsymbol{s}_i^p, \boldsymbol{s}_i^g\right)\right) + \frac{1}{N}\left(w_{\text{corr}}\mathcal{L}_{\text{corr}}\right) \quad (6)$$

where i refers to the index of the anchor and N is the total number of samples in a batch. $\boldsymbol{g}_i^*$ is the true label of the ith anchor, and $\boldsymbol{p}_i$ is the output of the classification branch in the detection header and indicates the probability value that its proposal belongs to the foreground. Here, w_{cle}, w_{reg}, and w_{corr} are the weights used for balancing. $\mathcal{L}_{\text{cls}}$ is the cross-entropy loss. $\mathcal{L}_{\text{reg}}$ is the smooth loss

$$\text{Smooth } L_1(x) = \begin{cases} 0.5\ x^2, & \text{if x} < 0 \\ |x| - 0.5, & \text{otherwise} \end{cases} \quad (7)$$

where $\boldsymbol{s}_i^p$ is the offset of the predicted bounding box with respect to the ith anchor and is a parameterized 5-D vector of the directed regression branch $\boldsymbol{s}_i^p = (s_x^p, s_y^p, s_w^p, s_h^p, s_\theta^p)$ (see Fig. 3)

$$s_x^p = \frac{x_p - x_a}{w_a}, \quad s_y^p = \frac{y_p - y_a}{h_a}$$
$$s_w^p = \log\frac{w_p}{w_a}, \quad s_h^p = \log\frac{h_p}{h_a}, \quad s_\theta^p = f\left(\frac{\theta_p - \theta_a}{\pi}\right) \quad (8)$$

where $\boldsymbol{b}_p = (x_p, y_p, w_p, h_p, \theta_p)$ and $\boldsymbol{b}_a = (x_a, y_a, w_a, h_a, \theta_a)$ denote the predicted bounding box of the object and the anchor box, respectively. $f(\cdot)$ is used to ensure that the angle difference does not exceed the preset range and to avoid the impact of angular periodicity. Similarly, the five normalized objects associated with the ground-truth bounding box and anchor box of the object are calculated as $\boldsymbol{s}_i^g = (s_x^g, s_y^g, s_w^g, s_h^g, s_\theta^g)$. Algorithm 1 summarizes the detailed procedure.

V. Experiments

This section mainly introduces the public datasets used for conducting object detection on remote sensing images, the performance evaluation metrics of the proposed algorithm, the experimental details, and the experimental results.

A. Datasets

DOTA [48] dataset is a large-scale remote sensing image dataset with a collection of 2806 aerial images. Each image is approximately 4000 × 4000 pixels in size and contains objects with various scales, orientations, and shapes. The DOTA images were annotated by remote sensing imagery experts for common instance classes. The fully annotated DOTA dataset contains 188 282 instances, and each is labeled by an oriented bounding box and a category. The DOTA dataset contains a total of 15 categories, including planes (PL), ships (SH), storage tanks (ST), baseball diamonds (BD), tennis courts (TC), basketball courts (BC), track-and-field grounds (GTF), harbors (HB), bridges (BR), large vehicles (LV), small vehicles (SV), helicopters (HC), roundabouts (RA), and soccer ball fields (SBFs). To ensure that the distributions of the training data and test data are approximately the same, the DOTA dataset is officially partitioned, and the training set, the test set, and the validation set consist of 1/2, 1/3, and 1/6 of the total data, respectively. Due to hardware resource limitations, we crop all 2806 images into slices with sizes of 1024 × 1024 pixels with a 200-pixel overlap; we finally obtain a total of 15 749 slices for the training set, 5729 slices for the validation set, and 10 833 slices for the testing set. Each slice also has its corresponding label. Finally, we compute the evaluation metrics and submit the results to the official evaluation website for 937 large raw test images.

HRSC2016 [54] is another dataset that is widely used for arbitrary ship direction detection; here, all images are acquired from six famous ports. It contains 1061 images with image resolutions ranging from 2 to 0.4 m and image sizes ranging from 300 × 300 to 1500 × 900, and most are scaled to 1000 × 600. The dataset includes three levels of tasks (single-class tasks, four categories, and 19 types of ship detection and recognition tasks) and is split into training, validation, and testing sets, which contain 436 images (including 1207 samples), 181 images (including 541 samples), and

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TABLE I
COMPARISON WITH STATE-OF-THE-ART METHODS ON DOTA DATASET

Methods	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
One-stage Methods																	
RetinaNet-O [21]	R-50-FPN	88.67	77.62	41.81	58.17	74.58	71.64	79.11	90.29	82.18	74.32	54.75	60.60	62.57	69.67	60.64	68.43
DAL [26]	R-101-FPN	88.61	79.69	46.27	70.37	65.89	76.10	78.53	90.84	79.98	78.41	58.71	62.02	69.23	71.32	60.65	71.78
RSDet [8]	R-152-FPN	90.10	82.00	53.80	68.50	70.20	78.70	73.60	91.20	87.10	84.70	64.30	68.20	66.10	69.30	63.70	72.10
R3Det [13]	R-152-FPN	89.49	81.17	50.53	66.10	70.92	78.66	78.21	90.81	85.26	84.23	61.81	63.77	68.16	69.83	67.17	73.74
S2A-Net [27]	R-50-FPN	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
R3Det-DCL [28]	R-152-FPN	89.78	83.95	52.63	69.70	76.84	81.26	87.30	90.81	84.67	85.27	63.50	64.16	68.96	68.79	65.45	75.54
Two-stage Methods																	
CAD-Net [30]	R-101-FPN	87.80	82.40	49.40	73.50	71.10	63.50	76.60	90.90	79.20	73.30	48.40	60.90	62.00	67.00	62.20	69.90
SCRDet [31]	R-101-FPN	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	**87.94**	86.86	65.02	66.68	66.25	68.24	65.21	72.61
FAOD [4]	R-101-FPN	**90.21**	79.58	45.49	76.41	73.18	68.27	79.56	90.83	83.40	84.68	53.40	65.42	74.17	69.69	64.86	73.28
RoI-Trans [32]	R-101-FPN	88.65	82.60	52.53	70.87	77.93	76.67	86.87	90.71	83.83	82.51	53.95	67.61	74.67	68.75	61.03	74.61
Gliding Vertex [33]	R-101-FPN	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
MaskOBB [34]	R-50-FPN	89.61	85.09	51.85	72.90	75.28	73.23	85.57	90.37	82.08	85.05	55.73	68.39	71.61	69.87	66.33	74.86
CenterMap [35]	R-50-FPN	88.88	81.24	53.15	60.65	78.62	66.55	78.10	88.83	77.80	83.61	49.36	66.19	72.10	72.36	58.70	71.74
ReDet [36]	R-50-ReFPN	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
Oriented RepPoints [38]	R-101-FPN	89.53	84.07	59.86	71.76	79.95	80.03	87.33	90.84	87.54	85.23	59.15	66.37	75.23	73.75	57.23	76.52
SCRDet++ [5]	R-152-FPN	88.68	85.22	54.70	73.71	71.92	84.14	79.39	90.82	87.04	86.02	**67.90**	60.86	74.52	70.76	**72.66**	76.56
FSDet [37]	R-50-FPN	89.20	84.33	55.38	74.81	80.85	78.79	88.24	90.85	86.85	85.85	67.14	64.79	76.49	75.11	69.02	77.85
DHRec [29]	R-152-FPN	88.95	**85.71**	**60.91**	76.01	80.57	84.13	88.75	90.89	85.48	**87.71**	64.32	70.86	77.22	73.26	67.70	78.83
DCDet	R-50-FPN	89.78	83.46	57.23	79.62	80.12	83.99	88.41	**90.90**	86.16	87.22	66.84	70.92	77.29	78.67	67.87	79.23
DCDet	R-101-FPN	89.61	85.05	55.63	81.31	80.57	**84.24**	88.55	90.89	85.71	87.21	66.10	71.63	77.46	77.86	67.71	79.30
DCDet	R-152-FPN	89.31	84.86	59.80	**82.93**	**81.19**	84.18	**88.97**	90.88	84.55	87.17	65.30	**72.59**	**79.14**	**79.22**	64.16	**79.62**

444 images (including 1228 samples), respectively. We only perform simple rotation and flipping on the images to prevent model overfitting.

DIOR-R [62] dataset is a large-scale benchmark dataset for object detection in optical remote sensing images. The dataset contains 23 463 images and 192 472 instances covering 20 object classes. These 20 object classes are airplane (c1), airport (c2), baseball field (c3), basketball court (c4), bridge (c5), chimney (c6), dam (c7), expressway service area (c8), expressway toll station (c9), golf course (c10), ground track field (c11), harbor (c12), overpass (c13), ship (c14), stadium (c15), storage tank (c16), tennis court (c17), train station (c18), vehicle (c19), and wind mill (c20). The dataset consists of 800 × 800 fixed-size images with a spatial resolution ranging from 0.5 to 30 m. There are a combined total of 11 725 images and 68 073 samples in the training and validation sets and 11 738 images and 124 445 samples in the testing set.

B. Experimental Details

We use PyTorch as our deep learning framework, and all experiments are conducted on a 32-GB NVIDIA RTX V100 GPU with a batch size of 8. The DOTA test results are evaluated on the official testbed. We use an R-CNN with an FPN as our baseline model, which utilizes Rotated RoIAlign for proposal feature aggregation. The ResNet backbone networks are pretrained on ImageNet. We jointly optimize the RPN and R-CNN heads to train our model (DCDet) in an end-to-end manner. Since a high degree of overlap is observed among the directed proposals generated by the RPN during the inference process and to attain reduced proposal redundancy, 2000 proposals are reserved for each of the five levels of the FPN in the first stage, and then, nonmaximum suppression (NMS) is applied [23]. We merge the proposals at each FPN level and select the proposals with the top 1000 classification scores as inputs for the second stage. The optimization method used for the model is the stochastic gradient descent (SGD) algorithm with a decay rate of 0.0001. We train the model for 12 epochs. The initial learning rate is set to 0.005 and divided by 10 at the eighth and 11th epochs. The ratio of training, validation, and test sets is 3:2:1 on the DOTA dataset, 1:1:2 on the DIOR-R dataset, and 2:1:1 on the HRSC2016 dataset, and we uniformly crop the divided images into slices of 1024 × 1024 size.

C. Comparison Results

We compare our DCDet method with 18 state-of-the-art remote sensing image object detection methods on the DOTA dataset, with 11 methods on the HRSC2016 dataset and with five methods on the DIOR-R dataset. Tables I–III show the comparison results obtained on both datasets. In the tables, R-50, R-101, and R-152 represent the use of ResNet-50, ResNet-101, and ResNet-152 as the backbone networks, respectively.

We compare the results obtained by the different state-of-the-art methods on the DOTA dataset in Table I, and the experimental results are uniformly evaluated on the official evaluation website [47], which is authentic and authoritative. From the experimental results, our DCDet method outperforms all of the compared methods. By utilizing ResNet-50, ResNet-101, and ResNet-152 as the backbone networks, DCDet obtains mAP values of 79.23%, 79.30%, and 79.62%. Both of these values are superior to those all of the compared methods. Fig. 6 shows some of the results obtained on the DOTA dataset. In Fig. 7, the performance curves on the validation set in the training process are compared, and the performance gap between the curves shows the effectiveness of each component. In Fig. 8, we show the comparison plot of the visualization effect after the SODS module.

For the HRSC2016 dataset, we list the 11 best mAPs achieved by the different methods, as shown in Table II. We conduct experiments on the HRSC2016 dataset using ResNet-50 as the backbone network of our model. Our method also achieves the best performance on this dataset with an mAP of 90.57%.

To further validate the effectiveness of our model, DCDet is compared with the state-of-the-art methods using the DIOR-R dataset. We evaluate DCDet on DIOR-R, and its results are compared to those of the state-of-the-art methods; these

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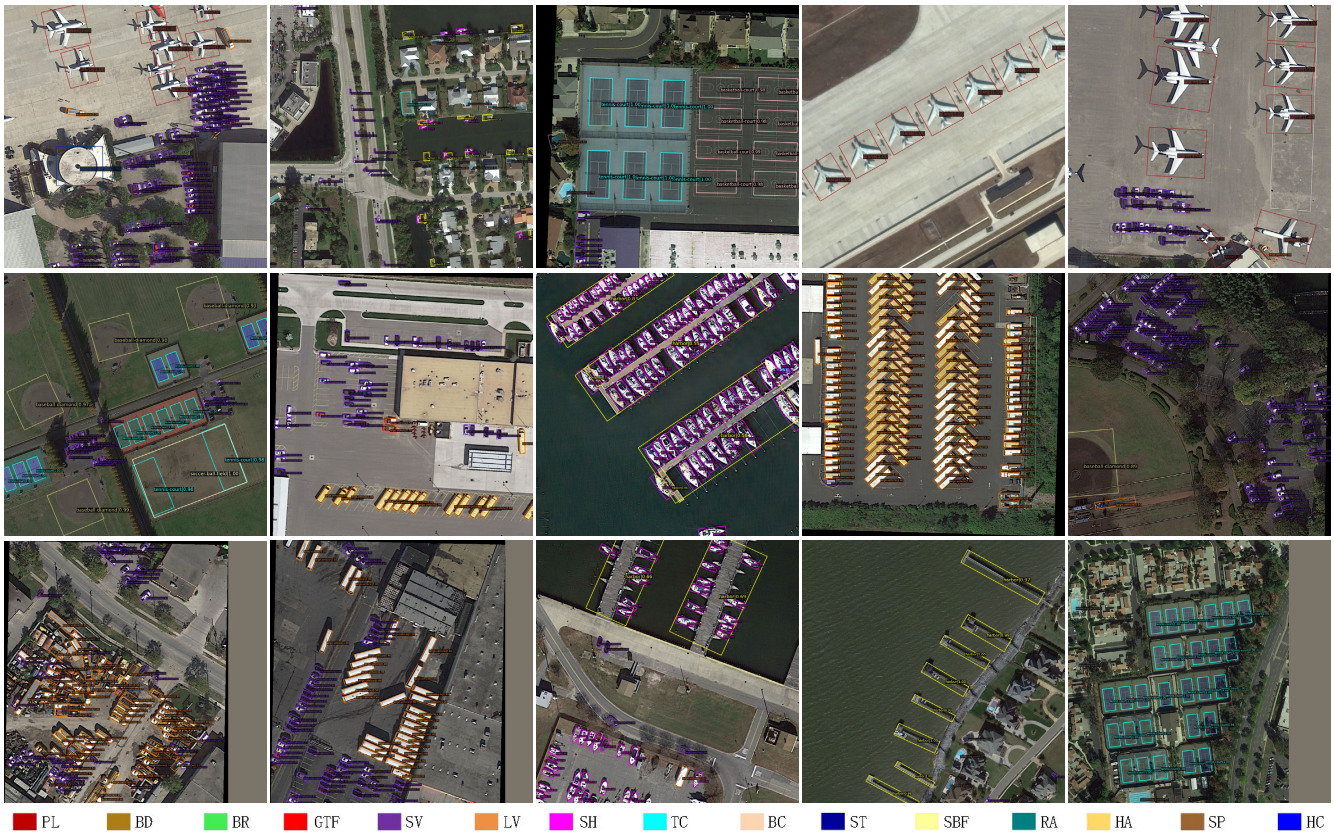


Fig. 6. Examples of detection results on the DOTA dataset.

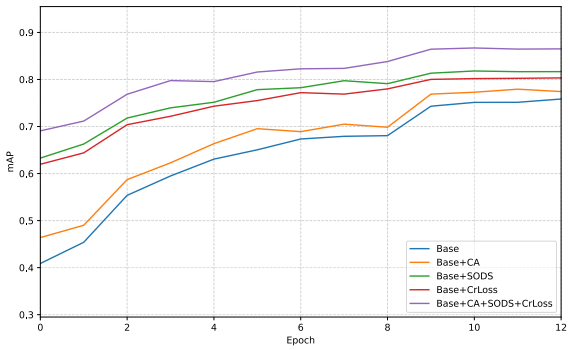


Fig. 7. Performance curves for the ablation experiments. We evaluate the performance for each training period on the DOTA validation set. The Y-axis shows the mAP50 for different modules with increasing training time.

methods include SRAF-Net [59], ASSD [9], RSADet [2], SB-MSN [60], and CoF-Net [61]. Our DCDet is the only method that exceeds 78%, as shown in Table III. DCDet achieves the best results in 11 out of 20 categories. These results show that our proposed method is able to effectively detect objects at different scales. Especially for categories containing a large number of small objects such as airplanes, ships, and vehicles, our DCDet has the best performance among all detectors.

D. Ablation Study

We conduct ablation experiments on the public DOTA dataset to analyze and determine the effectiveness of our DCDet method. The mAP metric is used to quantitatively evaluate the accuracy of the detection performance of DCDet. The mAP achieved on the DOTA dataset is determined by submitting the detection results to the DOTA evaluation website [48]. The main quantitative results are provided in Table IV and the network that we used is ResNet-152 as the feature extraction network. The quantitative results show that our proposed components can gradually improve the overall accuracy of our method.

Our baseline [55] model is a simple object detection network, where the backbone is ResNet pretrained on ImageNet. The input size is set to 1024 to ensure that the model can

TABLE II
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE HRSC2016 DATASET

Methods	Backbone	mAP
R2CNN [39]	ResNet-101	73.07
RRPN [42]	ResNet-101	79.08
R2PN [43]	VGG-16	79.60
RRD [40]	VGG-16	84.30
RoI Trans [32]	ResNet-101	86.20
Gliding Vertex [33]	ResNet-101	88.20
R3Det [13]	ResNet-101	89.26
DCL [28]	ResNet-152	89.46
KLD [41]	ResNet-101	89.87
Oriented R-CNN [55]	ResNet-101	90.50
ReDet [36]	ResNet-50	90.46
Ours	ResNet-50	**90.57**

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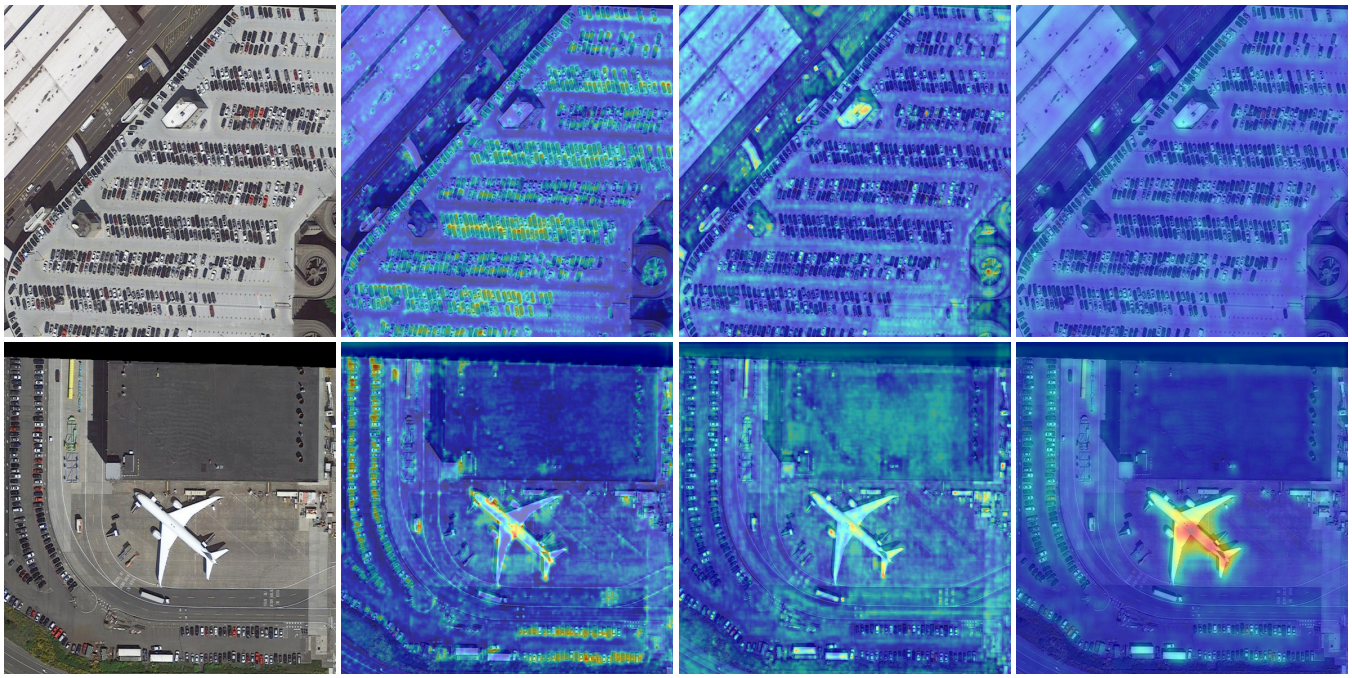


Fig. 8. Visualization comparison between our SODS and other feature fusion methods. It can be observed that our method exhibits significant advantages in detecting small objects while maintaining the detection performance for large objects.

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS ON DIOR-R DATASET

Methods				mAP	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10	c11	c12	c13	c14	c15	c16	c17	c18	c19	c20
SRAF-Net [59]				69.7	88.4	76.5	**92.6**	87.9	35.8	83.8	58.6	86.8	66.8	76.4	82.8	16.2	58.0	59.4	**80.9**	55.6	90.6	52.0	53.2	**91.0**
ASSD [9]				71.1	85.6	82.4	75.8	89.5	40.7	77.6	64.7	67.1	61.7	80.8	78.6	62.0	58.0	84.9	76.7	65.3	87.9	62.4	44.5	76.3
RSADet [2]				72.2	73.6	**86.0**	72.6	89.6	43.6	75.3	62.3	79.5	68.7	78.6	79.1	**57.9**	59.2	90.0	55.8	**77.0**	87.8	65.3	55.3	86.5
SB-MSN [60]				70.3	79.6	82.2	76.4	89.8	45.6	78.2	64.8	58.9	59.3	79.2	82.4	51.8	60.8	74.4	79.7	66.4	85.6	65.4	45.1	79.9
CoF-Net [61]				75.8	84.0	85.3	82.6	90.0	47.1	80.7	**73.3**	**89.3**	74.0	**84.5**	83.2	57.4	62.2	82.9	77.6	68.2	89.9	68.7	49.3	85.2
	CA	SODS	CrLoss																					
DCDet	✓			74.1	86.6	75.6	86.8	87.4	51.8	83.8	60.2	82.9	75.2	78.5	82.2	48.2	63.9	87.3	75.3	66.2	87.8	66.6	62.6	72.2
	✓	✓		76.7	89.2	72.8	85.7	89.4	56.8	84.8	60.0	88.1	77.5	78.3	86.4	51.1	**69.5**	90.2	80.6	73.7	90.8	68.4	64.5	76.2
	✓		✓	76.5	87.8	73.3	87.8	88.3	57.1	83.8	60.1	84.3	79.6	77.3	85.2	54.8	66.5	88.9	78.7	72.5	90.9	70.6	63.9	78.5
	✓	✓	✓	**78.3**	**90.1**	77.6	88.2	**90.6**	**58.0**	**85.9**	61.1	85.7	**80.6**	83.8	**87.7**	55.9	68.4	**91.3**	79.8	69.4	**91.7**	**72.9**	**65.2**	79.5

TABLE IV
OVERALL PERFORMANCE OF THE ABLATION EXPERIMENTS ON THE DOTA DATASET

	CA	SODS	CrLoss	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
	✓			89.08	83.61	54.59	71.49	78.96	77.95	87.32	90.89	80.52	86.33	64.61	68.73	76.05	70.54	58.36	75.93
DCDet (ResNet-152)	✓	✓		89.46	83.90	54.66	82.24	80.65	80.24	88.67	90.87	84.83	85.81	61.86	69.16	75.82	69.19	**68.18**	77.70
	✓		✓	**89.64**	84.53	57.61	77.69	80.08	83.92	88.49	**90.90**	**86.37**	86.93	63.24	69.58	76.83	71.88	67.04	78.31
	✓	✓	✓	89.31	**84.86**	**59.80**	**82.93**	**81.19**	**84.18**	**88.97**	90.88	84.55	**87.17**	**65.30**	**72.59**	**79.14**	**79.22**	64.16	**79.62**

better detect all remote sensing objects to build a more robust baseline model.

1) Effect of SODS: To verify the effectiveness of SODS, we compare the performances of models with and without this module in Tables III and IV. The results show that the overall detection accuracy (mAP50) of the model with SODS is improved by 1.77% on the DOTA dataset and 2.6% on the DIOR-R dataset, and the detection accuracy of small object categories is improving, which confirms that the SODS module can effectively alleviate the loss of small objects. In addition, we conduct further ablation experiments to investigate which scales of small object information are captured by SODS. Table V shows the results from this experiment. The experimental results demonstrate that regardless of which feature maps are sensed by SODS, they can enhance the performance of the model to some extent. The best performance is achieved when SODS dynamically senses the feature maps across the entire feature pyramid.

TABLE V
ABLATION STUDY ON DYNAMICALLY SENSING DIFFERENT COMPOSITIONS OF FEATURE MAPS

	$(P_2, P_3, P_4, P_5, P_6)$	(P_2, P_3, P_4, P_5)	(P_2, P_3, P_4)
mAP	76.78 $(+\mathbf{2.21})$	76.53 $(+\mathbf{1.96})$	76.39 $(+\mathbf{1.82})$

2) Effect of CrLoss: The performances of models with and without CrLoss are also compared in Tables III and IV. As observed in the results, the overall detection accuracy

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TABLE VI
CURRENT POPULAR DETECTORS COUPLED WITH OUR CRLOSS AFTER THE EFFECT OF ABLATION STUDY

	Method	mAP
One-Stage(+CrLoss)	R3Det [13]	73.74 (**74.59**)
	RetinaNet-O [21]	68.43 (**70.31**)
	S2A-Net [27]	74.12 (**75.05**)
	G-Rep [56]	78.47 (**78.97**)
Two-Stage(+CrLoss)	ReDet [36]	76.25 (**76.86**)
	Gliding Vertex [33]	75.02 (**76.58**)
	RoI-Trans [32]	74.61 (**76.10**)
	Oriented RepPoints [38]	77.63 (**78.31**)
	KFIoU [57]	75.18 (**76.42**)
	DCDet	77.70 (**79.46**)

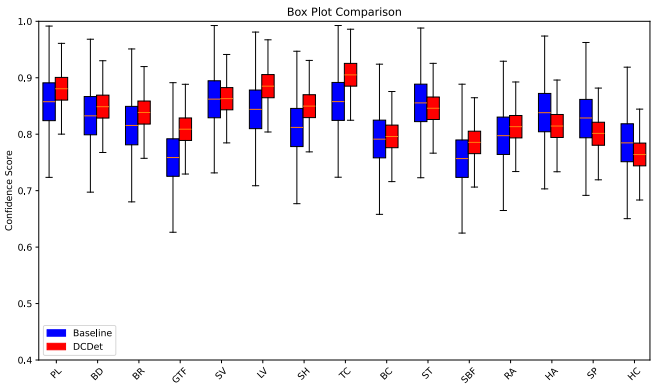


Fig. 9. Plot showing the confidence scores of DCDet with greater stability; these data provide convincing evidence for the robustness of DCDet.

of the model with CrLoss is improved by 2.38% on the DOTA dataset and 2.4% on the DIOR-R dataset. To further verify the effectiveness of our proposed CrLoss, we add it to other detection models, and the obtained results are shown in Table VI. Our CrLoss works well with the different detection models.

We conducted additional confidence interval experiments on DOTA to validate the reliability of DCDet. As shown in Fig. 9, we present a box plot showing the confidence scores of DCDet compared to the baseline [55], with the baseline model in blue and DCDet in red. The figure clearly demonstrates that the confidence scores of DCDet exhibit greater stability, providing convincing evidence for the robustness of DCDet.

To investigate the impact of our proposed module on the inference speed, we compare the inference speed (frames per second or FPS) of the different methods on the DOTA dataset, as shown in Table VII. The results show that the speed penalty from adding the CrLoss function module is minimal and negligible, but our mAP increases by 1.7% compared to 75.6% in the baseline [55]. When we add all modules to the inference speed, although there are some decreases in FPS, our mAP increases by 3.6% compared to the baseline, and we believe that the tradeoff is valuable.

We conducted DOTA tests on DCDet, and the results are presented in Table VIII. The performance of Baseline + CrLoss exhibits a 1.7% improvement, while the number of parameters and FLOPs remains relatively unchanged. These

TABLE VII
COMPARISON OF INFINITE SPEED ON DOTA (FPS)

Method	mAP	FPS
Faster RCNN	72.5	**16.0**
RoI Trans [32]	74.6	14.0
Gliding Vertex [33]	75.1	15.8
Baseline [55]	75.6	15.7
Baseline+CrLoss	77.3	15.1
DCDet	**79.2**	8.7

TABLE VIII
ABLATION EXPERIMENTS USING THE DOTA TEST SET

Method	mAP	Params	FLOPs	FPS
Baseline [55]	75.6	37.56M	6.6G	15.7
Baseline+CrLoss	77.3	39.25M	6.7G	15.1
Baseline+SODS	77.7	48.31M	7.4G	9.4
DCDet	79.2	49.95M	7.5G	8.7

results indicate that CrLoss has the capability to extract more resilient object features utilizing the same number of network parameters.

VI. CONCLUSION

In this article, we propose a new DCDet model for small object detection, which contains two main modules, SODS and CrLoss. SODS is extracted from the neck in the object detection model and uses 3-D convolution to process the pyramid feature maps in order to enhance the features of the intermediate layer of object detection. CrLoss is a simple and efficient CrLoss function that is a plug-in function that can be simply added to other detectors with classification and regression tasks to enhance the performance of the model. Extensive experimental results show that DCDet achieves state-of-the-art performance on the DOTA, HRSC2016, and DIOR-R datasets.

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